

Building a conceptual 3D geological model of the Wollaston fold-thrust belt, northern Saskatchewan: Implications for pegmatite-related rare earth elements (REE) targeting

Joseph Bravo^{1*}, Irvine R. Annesley^{1,2}, Camille A. Partin^{1**}¹University of Saskatchewan, Dept. of Geological Sciences, Saskatoon, Canada / ²Université de Lorraine - ENSG, Vandoeuvre-lès-Nancy, France

*joseph.bravo@usask.ca, **camille.partin@usask.ca

Introduction

Rare Earth Elements (REE) are essential for renewable technologies [1], but pegmatite-hosted REE systems remain difficult to explore because they are commonly small, structurally controlled [2], and poorly constrained in 3D [3, 4]. This PhD project focuses on integrated geological–geophysical modelling of the Wollaston Domain and forms part of an NSERC Alliance grant “Unlocking Canada’s Rare Earth Element (REE) Potential.” The Wollaston Domain is a prospective Precambrian terrane for U-Th-REE where regional lithotectonic architecture, structural corridors, and known REE occurrences provide a basis for improving regional-scale exploration models [5].

Objective

This study builds a regional-scale conceptual 3D geological model of the Wollaston Domain by integrating geological, geophysical, structural, and geochemical datasets to assess existing interpretations, identify knowledge gaps, and refine preliminary criteria for REE pegmatite

Research Hypotheses

H1: REE-bearing pegmatites likely reflects the interaction between fertile crustal source regions and inherited structural anisotropy.

H2: Northeast-trending D1/D2 ductile structures acted as melt–fluid pathways, whereas northwest-trending brittle faults (possibly late D3) influenced localization or preservation.

H3: Integrating lithostructural patterns with radiometric responses can help identify higher-priority targets for REE pegmatite exploration.

Geological Setting

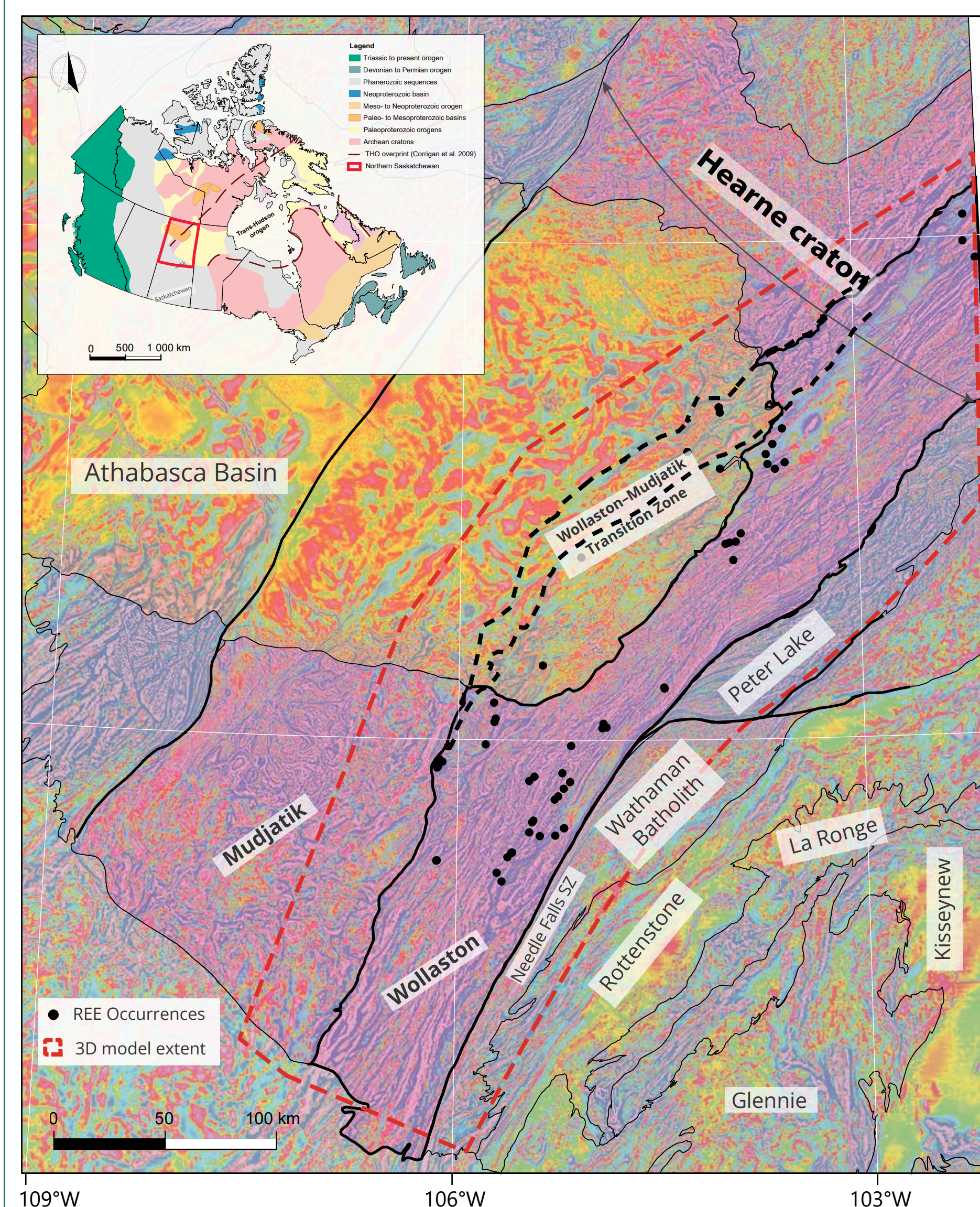


Figure 1. Location and geological–geophysical setting of the Wollaston Domain. The inset shows the study area within Canada and the Canadian Precambrian Shield. The main map shows lithotectonic domains of northern Saskatchewan [14] overlain on the first vertical derivative of total magnetic intensity (TMI) [15], with REE occurrences and the approximate 3D model extent [14].

- The Wollaston fold–thrust belt is a northeast-trending structural corridor within the Hearne craton that developed during Paleoproterozoic Trans-Hudson orogenesis [6–10].
- The belt records crustal shortening, partitioned strain, high-grade metamorphism, partial melting, and structural (transpressional and/or extensional) reactivation [4, 6, 9, 10].
- Archean basement and Paleoproterozoic Wollaston Group rocks are interfolded and intercalated within crustal-scale thrust sheets [6–10].
- The Wollaston-Mudjatik transition zone (WMTZ), an anastomosing network of basement shear zones and faults, separates basement-cored domes of the Mudjatik Domain from the linear fold–thrust architecture of the Wollaston Domain [4, 6, 12].
- Graphitic (in part carbonaceous) and highly deformed rocks in the retroshear WMTZ may have acted as heterogeneous fluid/melt pathways relevant to U and REE-bearing pegmatite systems [12, 13].

Data & Integration workflow

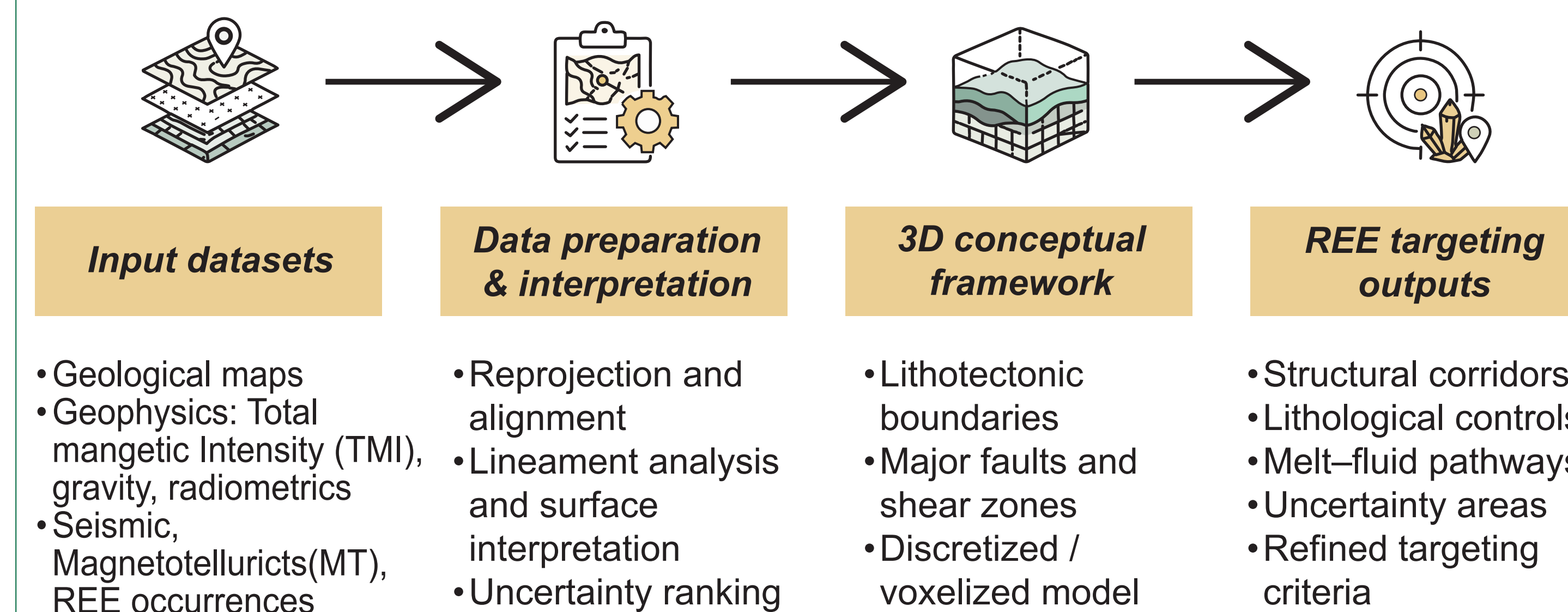


Figure 2. Integrated workflow used to build the 3D conceptual lithotectonic model. Surface geological and geophysical datasets are combined with subsurface seismic and MT constraints to evaluate structural architecture, uncertainty, and criteria relevant to REE pegmatite targeting.

3D Conceptual Model of the Wollaston Domain

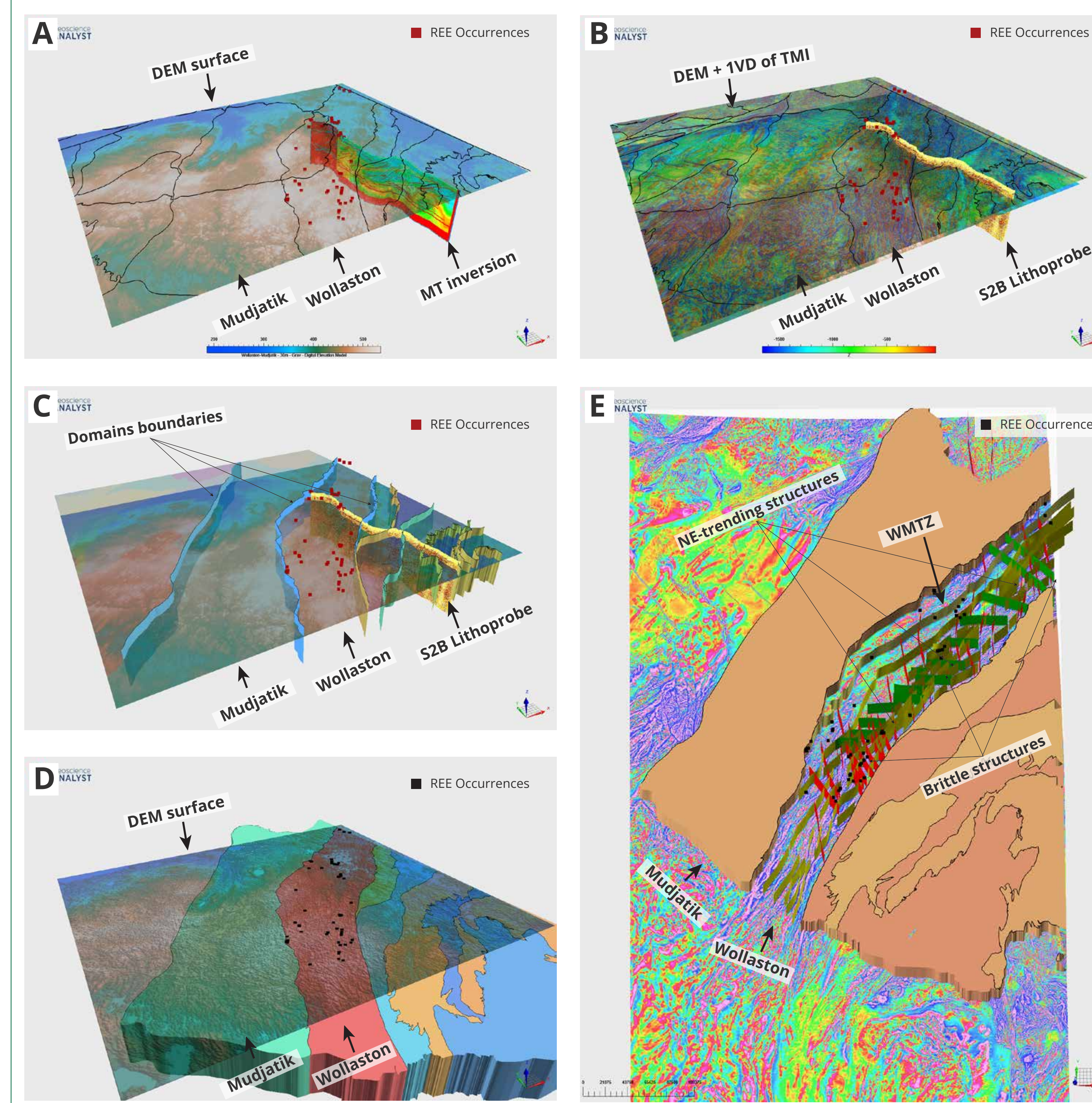


Figure 3. Preliminary 3D conceptual model inputs and derived structural framework. (A) topography/DEM surface [16] with MT inversion interpretation [17]. (B) Topography draped with the first vertical derivative of total magnetic intensity (TMI) and the Lithoprobe THOT S2B seismic transect [18], along which the MT profile was also acquired. (C) Interpreted vertical lithotectonic domain-boundary surfaces constrained by topography and showing its relationship with the S2B transect. (D) Preliminary lithotectonic block model of the Wollaston Domain. (E) interpreted structural model, with lineament-derived faults overlain on the first vertical derivative of TMI.

Key Findings

- Surface geology, geophysics, and subsurface constraints were integrated into a preliminary V1.0 conceptual model exploring the lithotectonic framework of the Wollaston fold-thrust belt.
- REE occurrences are spatially associated with regional to district-scale NE-trending structural corridors and interpreted shear-zone surfaces.
- NW-trending brittle structures may record late reactivation relevant to pegmatite localization and/or preservation.
- eTh radiometric patterns help evaluate preliminary links between REE occurrences, structures, and lithological domains.
- The model highlights uncertainty areas and refines preliminary criteria for REE pegmatite targeting.

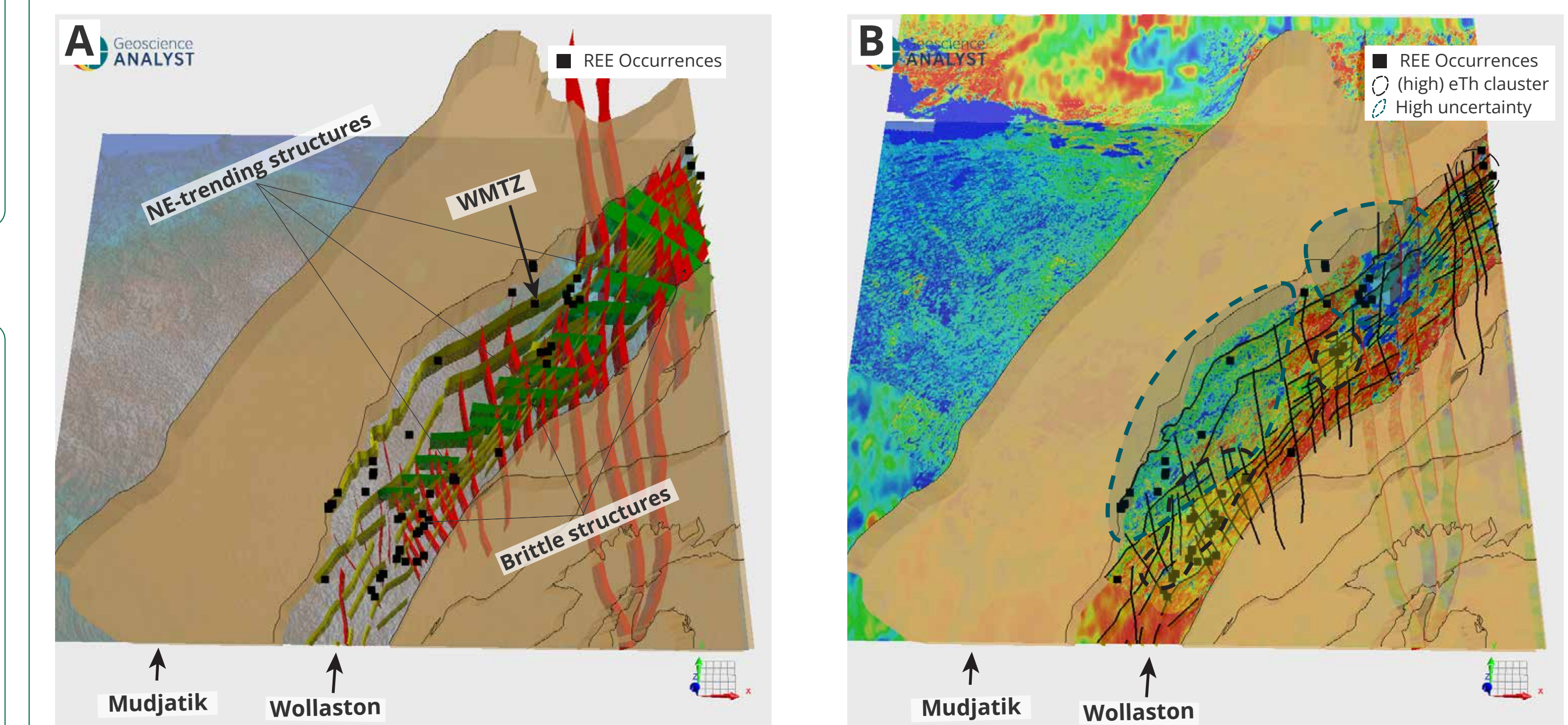


Figure 4. Preliminary spatial comparison between REE pegmatite occurrences, interpreted structures, and geophysical responses in the Wollaston Domain. (A) REE occurrences plotted with lineament-derived faults and shear-zone surfaces interpreted from the first vertical derivative of total magnetic intensity (TMI), highlighting the spatial relationship between pegmatite occurrences and NE–NW structural corridors. (B) Same structural framework overlain on equivalent thorium (eTh) radiometric data [15], showing the relationship between REE occurrences, interpreted structures, and radiometric response.

Future Work

- Refine the conceptual 3D lithotectonic model and uncertainty framework.
- Analyze and reinterpret Lithoprobe THOT Line S2B to better constrain subsurface geometry.
- Use diamond drill hole data in conjunction with tilt-derivative magnetics to constrain the 3rd dimension of the model (especially sub-Athabasca area of uncertainty).
- Use petrophysical data (including diamond drill hole data) for forward magnetic, gravity, and MT modelling.
- Perform geologically constrained inversions to iteratively improve the regional model.
- Evaluate and refine established criteria for preliminary REE pegmatite targeting.

Acknowledgements

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